

# Morning Edition

WEDNESDAY, 16 SEPTEMBER 2026

*Ten stories, filtered through SHORE Institute, ocean governance, and Shorelines.*

01 · Five Days to Sydney

02 · Five Million, Open Now

03 · 28.7 Percent

04 · Two Events, One Deadline Gone

05 · The Paradox, With Numbers

06 · Eight Years Since the Bond

07 · The Framework the Indian Ocean Hasn't Written

08 · The Heat Valve

09 · What HARMONiSE Is Becoming

10 · Forty Percent, Unpriced

**STORY 01**

 **ACT THIS WEEK**

## The Forum Deciding How Ocean Finance

# Gets Written, Five Days From Now

*The UN's Standing Committee on Finance meets in Sydney in five days to shape how climate money reaches water systems and oceans, and the SIDS financing gap you have spent years documenting is the forum's stated agenda item.*

## STORY NOTES

The 2026 Forum of the Standing Committee on Finance convenes 21 to 22 September at the Sofitel Sydney Darling Harbour, five days from today, on a single theme: Financing Climate Action for Water Systems and the Ocean. It is the first SCF forum to take ocean finance as its dedicated subject rather than folding it into a general climate finance agenda.

The background paper is blunt about the problem: ocean-based climate action is underfunded relative to country needs, against rising loss and damage, and against the conditional ocean commitments states have already written into their NDCs,

particularly for LDCs, SIDS, and vulnerable coastal nations. SIDS receive only a small share of total SDG 14 funding. AOSIS used the submission process to push for a technology repository or information access centre built specifically for SIDS scale and circumstances, delivered in partnership with regional organisations, close to the institutional model SHORE already argues for.

The written call for inputs closed in January, so that window is shut. What is still open is the forum itself: registration runs through Indico, and its outputs feed directly into the guidance the SCF gives the Green Climate Fund and Adaptation Fund boards over the following year.

**<https://indico.un.org/event/1025135/>**

#### STORY ARTICLE

Five days from today, the UN's Standing Committee on Finance opens a two-day forum in Sydney built around a single question: how does climate finance actually reach water systems and the ocean. That framing alone is worth noting. Ocean finance has spent most of its institutional life as a subsection of

broader climate finance discussions, competing for attention against mitigation infrastructure and land-based adaptation. This forum gives it the whole agenda.

The background paper prepared for the forum states the problem in terms you have made in Shorelines more than once: ocean-based climate action remains significantly underfunded relative to country needs, against rising loss and damage, and against the ocean commitments parties have already written into their nationally determined contributions, particularly for least developed countries, small island developing states, and other vulnerable coastal nations. The UNFCCC's own Standing Committee is describing the gap between commitment and delivery, not an outside advocacy group.

The submission record ahead of the forum is where this gets specific. AOSIS used its written input to argue for a technology repository or information access centre purpose-built for the scale and circumstances of small island states, delivered through partnership with regional organisations

rather than built fresh in each country. That is close to the institutional model HARMONiSE and the Ocean Mapping Hub already argue for at a Western Indian Ocean scale: shared regional infrastructure, not duplicated national capacity nobody can individually afford. Seeing that argument made at AOSIS level, inside a UN finance body's formal input process, is a stronger citation for that model than anything SHORE could produce making the same case alone.

Here is the counterargument worth naming before dismissing it: a two-day forum with no binding decision-making power is easy to treat as a talking shop, another convening that produces a summary report and changes nothing. That scepticism has some basis. SCF forums do not themselves allocate money. But they set terms other bodies with real money then apply. Forum outputs feed directly into the guidance the Standing Committee gives the Green Climate Fund and Adaptation Fund boards in the following funding cycle, and that guidance has shifted fund priorities before, most visibly around loss and damage framing after earlier forums.

The written call for inputs ahead of this forum closed in January, so that door is shut. What remains open is the forum itself, registration through the Indico portal linked above, and more usefully, the published outputs once the two days conclude. Those outputs are where the AOSIS technology repository proposal, and whatever direct SIDS ocean finance commitments emerge from the Sydney discussions, will be documented in language other institutions cite going forward.

The practical move is not to submit anything, that window is closed. It is to track the forum's published summary and background documentation once released, specifically for how the AOSIS technology repository idea is treated, and for any language on SIDS-specific ocean finance instruments a future Commonwealth or Shorelines argument can cite directly. A UN finance body's own words on the financing gap carry more weight in a funding pitch than the same argument made independently, however well evidenced.

<https://unfccc.int/event/2026-forum-of-the-standing-committee-on-finance>

STORY 02

 GRANT OPEN NOW

## ***Five Million Dollars, Open Now, No Deadline Posted Yet***

***The COP29 Presidency-Commonwealth Fund is handing out grants of up to two hundred thousand dollars to every Commonwealth SIDS government, Seychelles included, and applications are open with no closing date yet on the public record.***

### STORY NOTES

The COP29 Presidency-Commonwealth Fund for Small Island Developing States opened during London Climate Action Week in June 2026. It is a US\$5 million fund running over five years, and grants of up to US\$200,000 are available to the government of every Commonwealth SIDS, Seychelles among

them, with no competitive shortlist of eligible countries to clear first.

Eligible activities: early warning systems, climate risk planning, nature-based solutions, marine protection, sustainable fisheries, and resilient renewable energy infrastructure. That range covers a large share of what SHORE and Blue Economy-adjacent project proposals already argue for.

No confirmed application deadline appears in the public record as of this search. That is not the same as an open-ended window; funds like this typically close once allocated. Worth a direct enquiry to the Commonwealth Secretariat now rather than waiting for a public deadline announcement.

**<https://thecommonwealth.org/news/commonwealth-and-cop29-presidency-invite-commonwealth-sids-submit-climate-action-proposals>**

#### STORY ARTICLE

Here is a grant call with almost nothing standing between eligibility and applying. The COP29 Presidency-Commonwealth Fund for Small Island Developing States, launched jointly by the

Commonwealth Secretariat and Azerbaijan's COP29 Presidency at London Climate Action Week in June 2026, makes grants of up to US\$200,000 available to the government of any Commonwealth small island developing state. Seychelles qualifies by definition. There is no separate shortlisting round to clear before applying.

The fund's total size, US\$5 million across five years, is modest set against the financing gap figures SIDS advocates cite in every major forum. But modest and accessible is a useful combination for a specific kind of proposal: project design, capacity building, and pilot-scale implementation rather than full infrastructure financing. The eligible activities list reads like it was built from the SIDS ocean and climate policy agenda directly: early warning systems, climate risk planning, nature-based solutions, marine protection, sustainable fisheries, and resilient renewable energy infrastructure.

The stated purpose is to turn Commonwealth national climate plans into action, strengthening national capacity, improving policy and regulatory

frameworks, and unlocking further public and private investment through the practical partnership platforms the COP29 Presidency built. That framing, plans into action, capacity plus policy plus investment unlocking, maps closely onto what a HARMONiSE-adjacent proposal already needs funded: not the mapping infrastructure itself, but the institutional groundwork that makes future mapping investment fundable.

The gap in the public record is the deadline. Every source describing this fund states the grant ceiling and the eligible activities clearly, and none states when applications close. That could mean the fund runs on a rolling basis until the five-year allocation is exhausted, or it could mean a deadline exists that simply has not surfaced in searchable coverage yet. Assuming the more generous interpretation and waiting for confirmation is the riskier of the two options.

The practical response to an unconfirmed deadline on an open, eligible grant is not to wait for the deadline to appear. It is to treat the absence of a

public deadline as reason to move now rather than as evidence there is no rush. A direct enquiry to the Commonwealth Secretariat's climate finance desk, confirming application mechanics and any internal timeline, costs an email and removes the guesswork.

Weigh this against the Commonwealth Blue Charter Project Incubator covered in an earlier edition, a smaller, slower-moving fund Seychelles has already drawn on once. The COP29 Presidency fund is newer, larger per grant, and explicitly ocean-and-climate focused rather than blue-economy-general. The two are not competing for the same proposal. A SHORE-adjacent early warning or marine protection pilot fits this fund's stated activities more directly than it fits the Incubator's project-development framing, and that distinction is worth having clear before drafting anything.

**<https://thecommonwealth.org/news/commonwealth-si-and-cop29-presidency-invite-commonwealth-sids-submit-climate-action-proposals>**

## 28.7 Percent

*Every fraction of a percentage point the GEBCO grid adds is baseline data the Ocean Mapping Hub does not have to generate alone, and the region that just tripled its coverage did it by pooling data everyone already had.*

# 28.7%

OF THE WORLD'S OCEAN FLOOR NOW MAPPED TO  
MODERN STANDARDS — 104 MILLION KM<sup>2</sup>, UP FROM 27.3%  
A YEAR AGO

### STORY NOTES

The 2026 release of the GEBCO grid puts 28.7% of the world's ocean floor mapped to modern standards, up from 27.3% the year before. That is 104 million square kilometres in total, with close to 5 million square kilometres added over the past twelve months.

The regional detail matters more than the global figure. The ROPME Sea Area, essentially the Gulf, more than tripled its coverage in a single cycle, from 6.4% to 20.5%. That jump came from pooling existing survey, industry, and crowdsourced data rather than commissioning new expeditions.

The Western Indian Ocean has no equivalent story in this release. At the current global pace, full coverage by the 2030 target would require roughly five times the present annual mapping rate, sustained every year from here, a pace nothing in this release suggests is coming.

**<https://www.gebco.net/news/release-gebco2026-grid>**

#### STORY ARTICLE

Twenty eight point seven percent. That is the whole of the ocean floor mapped to modern bathymetric standards, worldwide, according to the 2026 release of the GEBCO grid from the Nippon Foundation-Seabed 2030 partnership. A year ago the figure stood at 27.3%. Run that pace forward and the 2030 full-coverage target is not on track, not by a wide

margin. Closing the remaining 71.3% in four years would need something close to five times the current annual mapping rate, sustained every year without a funding gap. Nothing in this release suggests that acceleration is coming.

The regional breakdown is the more useful number to sit with. The ROPME Sea Area, the Gulf, jumped from 6.4% coverage to 20.5% in a single cycle, more than tripling. That did not happen through a dedicated new mapping campaign. It happened because states in the region, industry operators already running seismic and geotechnical surveys, and crowdsourced bathymetry contributors pooled data that mostly already existed, and Seabed 2030 counted the aggregate result. That is a template worth naming directly: coordinated data-sharing among actors who already have vessels and instruments in the water beats waiting on new dedicated expeditions that competing budget priorities will keep delaying.

The Western Indian Ocean has no comparable jump in this release. Seychelles's exclusive economic zone runs to roughly 1.4 million square kilometres, one of

the largest EEZ-to-landmass ratios on the planet, sitting mostly inside a region where modern-standard bathymetric coverage remains thin. That gap is not a criticism of any single actor in the region. It is precisely the problem HARMONiSE and the Ocean Mapping Hub exist to address, and this annual release is the scorecard the Hub's own progress should be measured against. A future Hub progress report that cannot point to a coverage jump resembling the ROPME figure, even at a smaller scale, needs a harder answer than a general appeal to more mapping being needed.

There is a second use for this data beyond the scorecard function. Now that BBNJ is in force, area-based management tools for the high seas, meaning the legal mechanism for designating marine protected areas beyond national jurisdiction, require credible baseline bathymetric and ecological data to support any nomination. A coverage gap in the Indian Ocean is therefore not only a mapping problem. It is a hard constraint on what Seychelles or any Indian Ocean coalition can credibly propose once BBNJ's

Conference of the Parties opens that nomination process.

The ROPME template deserves closer study than a single paragraph gives it. What exactly got pooled, whose data counted, and under what data-sharing terms, is the more interesting question than the headline percentage. That detail, once published in full, is the material worth pulling directly for any Western Indian Ocean data-sharing proposal modelled on what just worked in the Gulf.

<https://www.gebco.net/news/release-gebco2026-grid>

04



CALENDAR + CONTACT

## Two Seabed 2030 Events, One Deadline Already Gone

*Two Seabed 2030 gatherings land on the calendar within two weeks of each other this*

## ***October, and the travel-funded registration window for the one closest to Seychelles has already shut.***

### STORY NOTES

Two Seabed 2030 events follow directly from this year's GEBCO grid release. The GEBCO Symposium runs in Cartagena, Colombia, on 8 and 9 October, bringing together the global ocean community to exchange knowledge and advance collective understanding of the seabed and ocean environment. The Seabed 2030 Indo-Pacific Ocean Mapping Meeting and Data Sprint runs in Kuala Lumpur, Malaysia, from 20 to 22 October, aimed at hydrographic offices,

### STORY ARTICLE

Two Seabed 2030 gatherings land within two weeks of each other this October, and reading them together is more useful than reading either alone. The GEBCO Symposium convenes in Cartagena, Colombia, on 8 and 9 October, the traditional venue for the ocean mapping community's broader knowledge exchange. The Seabed 2030 Indo-Pacific Ocean Mapping Meeting and Data Sprint follows in Kuala Lumpur, Malaysia, from 20 to 22

academic institutions, research organisations, industry, and government agencies across the Indo-Pacific region.

The deadline for in-person registration and travel fund requests for the Kuala Lumpur meeting closed on 12 August, so that specific door is shut for this cycle. Requests for an invitation letter still route through [pacific@seabed2030.org](mailto:pacific@seabed2030.org).

What remains genuinely open is the published output of both events once they conclude, particularly the Kuala Lumpur data sprint's methodology, since that

October, built specifically around hydrographic offices, academic institutions, research organisations, industry, and government agencies working across the Indo-Pacific.

The Kuala Lumpur meeting is the one worth the closer look, because it sits closest in both geography and mandate to the coverage gap the Western Indian Ocean is running against. Its stated purpose is to strengthen coordination around ocean mapping across the Indo-Pacific and to accelerate contributions to the GEBCO grid through a dedicated data

meeting sits closest geographically and thematically to the Western Indian Ocean's own coverage gap.

sprint, a practical, collaborative exercise in preparing and integrating existing data rather than commissioning new surveys. That is the same mechanism that tripled the ROPME Sea Area's coverage this year, applied to a different region.

The deadline that has already passed matters here, and it is worth stating plainly rather than treating the event as still fully open. In-person registration and travel fund requests for the Kuala Lumpur meeting closed on 12 August. That window is shut for this cycle. Anyone hoping to

attend in person on Seabed 2030 travel support needed to have applied over a month before this edition went to print.

What that closed window does not foreclose is everything downstream of the meeting itself. Data sprint methodologies, once published, are typically documented well enough to be studied and adapted without having been in the room. A Western Indian Ocean equivalent of the Kuala Lumpur data sprint, run at a smaller scale with regional hydrographic offices and whatever industry survey data

already exists in the region, is a concrete, costed pitch the Ocean Mapping Hub could make once the Kuala Lumpur methodology is public, citing it directly as precedent.

The Cartagena symposium is the broader convening and less directly actionable, but worth tracking for the same reason any global ocean mapping meeting is worth tracking: it is where funding priorities and coverage targets for the 2030 deadline get discussed at the level that eventually filters down into which regions

get prioritised for support.

The near-term move is not registration, that ship has sailed for Kuala Lumpur this cycle. It is a direct email to [pacific@seabed2030.org](mailto:pacific@seabed2030.org), both to ask whether any late or virtual participation options remain open and to get on record early for the next cycle, or more usefully, for a future Western Indian Ocean equivalent meeting, should one exist.

**<https://seabed2030.org/2026/06/19/seabed-2030-indo-pacific-ocean-mapping-meeting/>**

# The Paradox, With This Year's Numbers Attached

*Seychelles just produced a textbook year of the High-Income SIDS Paradox: record growth, record debt, and a single geopolitical shock away from both reversing at once.*

## STORY NOTES

Seychelles closed 2025 with record tourism arrivals of 398,841 visitors and real GDP growth of 5.8%, according to World Bank figures, the kind of headline number that reads as unambiguous success. Public and publicly guaranteed debt told a different story in the same period, rising from 53.6% of GDP at the end of 2025 to a projected 57.2% by the end of 2026, including new guarantees.

President Patrick Herminie took office on 26 October 2025 after defeating incumbent Wavel Ramkalawan

by a narrow 52.7% in a runoff, inheriting both figures at once. Fitch now projects a 10% drop in 2026 tourist arrivals and an 8% fall in tourism revenue, attributing the reversal to disruption from the US-Iran war rather than anything Seychelles's government controls.

Seychelles holds Africa's highest per capita income and, by that measure alone, reads to most financing instruments as a country that does not need concessional support. The debt trajectory and the tourism dependency say otherwise.

**<https://www.riotimesonline.com/seychelles-economy-tourism-blue-bonds-2026/>**

#### STORY ARTICLE

**S**eychelles just handed the High-Income SIDS Paradox argument a single-country case study with this year's numbers attached, and the numbers argue the point better than the framework alone ever could. Record 2025 tourism arrivals of 398,841 visitors pushed real GDP growth to 5.8%, according to World Bank figures, a headline any finance ministry would want to lead with. In the same period, public

and publicly guaranteed debt rose from 53.6% of GDP at the end of 2025 to a projected 57.2% by the end of 2026, once new guarantees are counted.

Those two figures sitting side by side are the paradox in miniature. Growth looks strong enough that Seychelles reads, to most external financing screens, as a country managing its own affairs competently and therefore not a priority for concessional support. The debt trajectory underneath that growth tells a narrower story: an economy expanding on the back of a single sector, tourism, financed partly through guarantees that do not disappear if that sector contracts.

That contraction is not hypothetical. Fitch now projects a 10% drop in 2026 tourist arrivals and an 8% fall in tourism revenue, attributed to disruption from the US-Iran war rather than any domestic policy failure. Objection worth naming: a single geopolitical shock unconnected to domestic governance looks like thin evidence for a structural argument. But that is exactly the mechanism the High-Income SIDS Paradox describes. A structurally narrow economy

does not need a shock tied to its own choices to take the hit. Any sufficiently large external disruption to its single dominant sector will do, and there will always be another one.

President Patrick Herminie inherited both the record and the reversal at once, taking office on 26 October 2025 after a runoff decided by 52.7% of the vote, a margin thin enough that governing capacity itself is not a given. That is the political backdrop against which a 3.6 percentage point jump in debt-to-GDP in a single year has to be managed, alongside a tourism sector now projected to contract rather than repeat 2025's record.

Africa's highest per capita income is the figure that keeps Seychelles locked out of the concessional windows, readiness funding, and LDC-targeted instruments that lower-income SIDS can still draw on, the exact mechanism the Back to Blue \$12 billion financing gap report, covered in an earlier edition, describes at a global scale. GNI per capita does not measure debt trajectory, sector concentration, or exposure to shocks with no domestic cause. This

year's figures are the concrete, dated evidence that the gap between what the income classification measures and what the country's actual fiscal exposure looks like is not a theoretical mismatch. It is a live one, unfolding in the same twelve months as the record year that supposedly proved Seychelles did not need help.

The useful move here is citation, not alarm. A Shorelines piece or funding proposal arguing the High-Income SIDS Paradox now has a specific, sourced, current-year data point to anchor the argument rather than gesturing at it in general terms: 5.8% growth, 398,841 arrivals, 53.6% to 57.2% debt-to-GDP, and a Fitch-projected reversal, all inside the same eighteen months.

---

Source: Rio Times Online, "Seychelles Economy Slows as President Herminie Inherits Record Tourism Boom and Rising Debt," 2026, citing World Bank and Fitch figures.

# Eight Years Since the Bond That Started All of This

*The world's first sovereign blue bond turns eight this October, and its actual mechanics, not the headline, are the model worth re-reading before the next instrument gets pitched.*

## STORY NOTES

Seychelles issued the world's first sovereign blue bond on 29 October 2018, raising US\$15 million. The World Bank structured the deal with a US\$5 million guarantee and a US\$5 million concessional loan from the Global Environment Facility, partially covering interest payments and making the bond investable at a scale ordinary sovereign debt from a small economy rarely achieves. Calvert Impact Capital, Nuveen, and Prudential Financial were the investors.

The bond followed, and built directly on, a 2014 debt-for-nature swap: roughly US\$21.6 million in debt

restructured with Paris Club creditors and The Nature Conservancy, which created SeyCCAT, the Seychelles Conservation and Climate Adaptation Trust, as the standing mechanism for channelling both instruments' proceeds.

Proceeds from the blue bond went toward expanding marine protected areas to 30% of Seychelles's exclusive economic zone, improving fisheries governance, and developing blue economy sectors including ecotourism and seafood value chains, managed jointly through SeyCCAT and the Development Bank of Seychelles.

**<https://www.worldbank.org/en/news/press-release/2018/10/29/seychelles-launches-worlds-first-sovereign-blue-bond>**

#### STORY ARTICLE

The world's first sovereign blue bond turns eight this October, and the anniversary is a better prompt to re-read the mechanics than to repeat the headline. Seychelles issued the bond on 29 October 2018, raising US\$15 million, a modest sum against most sovereign debt instruments but not modest against

what it was built to prove: that a small island economy could access capital markets for ocean conservation on terms an investor would actually take.

The structure is the part worth studying closely, because the number alone does not explain why it worked. The World Bank did not simply arrange the bond. It backed it with a US\$5 million guarantee and paired that with a US\$5 million concessional loan from the Global Environment Facility, structured to partially cover interest payments. That blended layering, guarantee plus concessional interest support plus the sovereign issuance itself, is what turned a small, unproven instrument from an untested small state into something Calvert Impact Capital, Nuveen, and Prudential Financial were willing to buy. Strip out either the guarantee or the concessional interest cover and the same bond likely does not clear at investable terms.

The bond did not appear from nothing. It built directly on a 2014 debt-for-nature swap, roughly US\$21.6 million in Seychelles debt restructured with Paris Club

creditors and The Nature Conservancy. That swap is what created SeyCCAT, the Seychelles Conservation and Climate Adaptation Trust, the institution that then became the standing channel for the blue bond's proceeds four years later. The sequencing matters: the institutional plumbing, a functioning trust with a track record of managing conservation finance, existed before the bond needed it, rather than being built in parallel with the bond negotiation itself.

Proceeds went toward expanding marine protected areas to 30% of the exclusive economic zone, improving fisheries governance, and developing blue economy sectors, ecotourism and seafood value chains specifically, managed jointly through SeyCCAT and the Development Bank of Seychelles. That MPA expansion target is not an abstraction anymore. It is the same 30% figure now echoed in the Kunming-Montreal Global Biodiversity Framework's 30×30 target, which Pacific states are formally working toward through the framework adopted at PCNC11, covered elsewhere in this edition. Seychelles hit its version of that number years before it became the global standard.

One fair challenge here: blue bonds have not scaled the way their 2018 debut promised. Few other small states have replicated the exact structure at comparable terms, and the instrument remains more a proof of concept than a template in wide use. That criticism holds on volume. It says less about the mechanics themselves, which is where the real lesson sits. The blend of guarantee, concessional interest cover, and a pre-existing trust institution is a specific, reproducible formula, not a one-off. What has not been solved yet is which other small states have the institutional plumbing, a SeyCCAT equivalent, ready before their own instrument gets pitched to investors.

Eight years on, the bond's real legacy is that formula, not the fifteen million dollar headline number. Any future Seychelles or Western Indian Ocean instrument pitched on the back of this precedent needs the guarantee-plus-concessional-plus-institution structure explicit in the pitch, not the achievement alone.

<https://www.worldbank.org/en/news/press-release/2018/10/29/seychelles-launches-worlds-first-sovereign-blue-bond>

STORY 07

## The Pacific Just Adopted the Framework the Indian Ocean Hasn't Written

*The Pacific just adopted a five-year regional conservation framework with a named finance stream inside it, and no equivalent document exists yet for the Western Indian Ocean SIDS you argue for.*

### STORY NOTES

The 11th Pacific Islands Conference on Nature Conservation and Protected Areas ran 7 to

### STORY ARTICLE

The Pacific just did something the Western Indian Ocean SIDS you write for have not:

11 September at the Tjibaou Cultural Centre in Noumea, New Caledonia, convened by the Government of New Caledonia, SPREP, and the Pacific Islands Roundtable for Nature Conservation. Delegates adopted the Pacific Regional Framework for Nature Conservation 2026-2030 and the State of Environment and Conservation 2025 report.

The conference programme ran across six thematic streams mirroring the framework's pillars, including area-based conservation and

adopted a five-year regional conservation framework with conservation finance named explicitly as one of its governing streams. The 11th Pacific Islands Conference on Nature Conservation and Protected Areas ran from 7 to 11 September at the Tjibaou Cultural Centre in Noumea, New Caledonia, convened jointly by the Government of New Caledonia, SPREP, and the Pacific Islands Roundtable for Nature Conservation. Delegates adopted the Pacific Regional Framework for Nature Conservation 2026-2030 alongside the

connectivity toward the 30×30 target, species and ecosystem recovery, invasive species prevention, nature-based solutions for climate adaptation and disaster resilience, and environmental governance and leadership, the stream explicitly naming conservation finance.

A Pacific youth declaration issued alongside the conference on 11 September called for integrated action across biodiversity, climate change, and ocean action, positioning the next generation's demands directly inside

State of Environment and Conservation 2025 report and a new declaration.

The conference structure is worth studying as closely as its outcomes. Six thematic streams organised the programme, mirroring the framework's own pillars: area-based conservation and connectivity, tracking Pacific progress toward the Kunming-Montreal Global Biodiversity Framework's 30×30 target across terrestrial, marine, and freshwater protected areas; species and ecosystem recovery; invasive species

the region's formal framework rather than as a parallel statement.

prevention and biosecurity; nature-based solutions for climate adaptation, disaster resilience, and food and water security; and environmental governance and leadership, the stream that names conservation finance directly rather than treating it as an implied resourcing question sitting outside the framework's scope.

That last point is the one worth sitting with. Most regional conservation frameworks treat finance as an implementation footnote, something to be sorted out after the substantive targets are

agreed. The Pacific framework builds conservation finance into a named governing stream alongside species recovery and invasive species control, which changes how the finance conversation gets structured going forward: not a separate fundraising exercise bolted onto agreed targets, but a governance question with the same standing as the ecological ones.

The Western Indian Ocean has nothing equivalent. Seychelles, Comoros, Mauritius, Madagascar, and the Maldives share more of

the Pacific SIDS's structural profile, small population, narrow economic base, high climate exposure, acute biodiversity value, than they share with most large mainland states, yet there is no standing regional framework doing for this group what the Pacific Regional Framework for Nature Conservation now does for its region: a shared five-year document with named streams, a finance mandate built in, and a formal adoption process behind it rather than a set of parallel national strategies.

A Pacific youth declaration issued alongside the conference on 11 September adds a second layer worth noting: it called for integrated action across biodiversity, climate change, and ocean action, and it did so inside the formal conference process rather than as a separate advocacy statement issued afterward.

Embedding the youth voice in the adoption process itself, rather than treating it as commentary on a document already finalised, is a structural choice other regions convening similar

frameworks have not consistently made.

This is a template, not just a comparison point. SHORE sits in a position to propose the Western Indian Ocean equivalent process, explicitly modelled on the Pacific framework's structure, its six streams, its named finance mandate, and its youth integration, rather than starting the pitch from a blank page. Citing a freshly adopted, comparable regional precedent is a stronger opening position than arguing the need for a framework in the abstract.

# The Atlantic Current Everyone Thought Just Moved Heat Around Actually Traps It

*New paleoclimate work shows the Atlantic circulation everyone modelled as a heat redistributor is actually a valve that traps and releases heat for the whole planet, and a weakening AMOC now reads as an accelerant, not just a regional cooling risk.*

---

*"The AMOC works like a heat valve that controls the energy budget of the planet."*

## STORY NOTES

A study led by paleoclimatologist Christo Buizert, published in *Nature Geoscience*, examined abrupt reorganisations of the Atlantic Meridional Overturning Circulation during the last ice age and found the standard explanatory model wrong in a specific, correctable way. The conventional "thermal bipolar seesaw" framework treats AMOC slowdowns as redistributing heat between hemispheres. Buizert's team found the ocean's total heat content actually increases during weak AMOC phases and is released during strong phases, meaning the AMOC functions as a valve controlling how much heat escapes the planet, not merely where existing heat moves.

The AMOC has stayed strong for the roughly 11,700 years since the last ice age ended, but most climate models project future weakening under continued warming. Buizert's framework means that projected weakening reads as a mechanism trapping additional heat inside the planetary system, amplifying

warming, rather than only the more commonly cited regional cooling risk to Western Europe.

This reframes what an AMOC slowdown means for tropical and equatorial ocean regions, including the Indian Ocean, where the usual cooling-risk narrative barely applies but the heat-accumulation mechanism applies everywhere.

**<https://phys.org/news/2026-08-weakening-atlantic-ocean-current-earth.html>**

#### STORY ARTICLE

For as long as AMOC weakening has been a mainstream climate risk discussion, the dominant story has been about Western Europe losing the warm water that keeps its winters mild. A new paleoclimate study led by Christo Buizert at Oregon State University, published in Nature Geoscience, argues that framing has always undersold the actual planetary stakes, and it does so using ice age data rather than future model projections.

Buizert's team examined abrupt AMOC reorganisations during the last glacial period, events well documented in ice cores and ocean sediment

records, and tested them against the conventional explanatory model, the thermal bipolar seesaw, which describes AMOC slowdowns as redistributing existing heat between the northern and southern hemispheres. Slow the circulation and the north cools while the south warms, heat moving around a system whose total content stays roughly fixed. That is the model most climate communication still leans on when explaining AMOC risk.

The finding that overturns this: the global ocean's total heat content does not stay fixed through these events. It increases during weak AMOC phases and is released during strong phases. In Buizert's own framing, the AMOC works like a heat valve that controls the energy budget of the planet, not a heat conveyor redistributing a constant total. When the valve is open, heat drains out of the ocean system. When it closes, heat accumulates inside it.

The mechanism runs through radiative feedbacks tied to how much heat escapes to the atmosphere and space depending on circulation strength, not simple redistribution. That matters because the

AMOC has held strong throughout the roughly 11,700 years since the end of the last ice age, the entire span of stable climate human civilisation developed inside. Most current climate models project future AMOC weakening under continued warming. Read through the redistribution model, that projected weakening was a regional risk, mainly to the North Atlantic and Western Europe. Read through Buizert's heat valve model, the same projected weakening reads as a mechanism that traps additional heat inside the whole planetary system, amplifying global warming rather than simply relocating its effects.

This is where the finding stops being a North Atlantic story and becomes an ocean-wide one. The conventional AMOC narrative has never had much to say to equatorial and tropical ocean regions, including the Indian Ocean, because the cooling risk it describes is geographically specific to the North Atlantic basin. A heat-trapping valve mechanism has no such geographic limit. If a weakening AMOC accumulates heat across the global ocean system rather than shifting it regionally, that accumulated heat does not stay confined to the basin where the

circulation change originated. It becomes a planetary energy budget question, directly relevant to sea surface temperature and marine heatwave trends in the Western Indian Ocean, even though the AMOC itself sits an ocean away.

The practical takeaway for anyone working ocean climate risk outside the North Atlantic is to stop treating AMOC weakening coverage as a story about someone else's regional cooling. Buizert's reframing makes it a global heat budget mechanism, and that is the version of the finding worth citing when the next Shorelines piece needs to connect distant circulation changes to warming trends closer to home.

**<https://phys.org/news/2026-08-weakening-atlantic-ocean-current-earth.html>**

---

STORY · 09

## **The Paper That Already Named What HARMONiSE Is**

# Becoming

*A governance framework already exists naming what happens when a data initiative like HARMONiSE stops being a data provider and starts being a governance actor in its own right, and it is worth citing by name.*

## STORY NOTES

A framework published through Earth Systems Governance research, analysing what it calls Ocean Data Science Initiatives, catalogues more than 150 global and regional initiatives that mobilise geospatial and temporal ocean data with the explicit goal of improving oceans governance, not just describing it.

The framework's central claim: these initiatives are creating novel forms of agency, becoming significant ocean governance actors in their own right, and in doing so are blurring the previously clear boundary between governance architecture, the formal institutions, treaties, and bodies, and governance agency, who actually exercises influence over outcomes.

The International Seabed Authority posed a version of this same question publicly this year, asking whether 2026 is the point at which ocean governance architecture needs formal consolidation to keep pace with exactly this kind of initiative-driven authority.

**<https://www.sciencedirect.com/science/article/pii/S258981162200009X>**

#### STORY ARTICLE

There is a framework that already has a name for what HARMONiSE is in the process of becoming, and it predates HARMONiSE by several years. Research applying an Earth Systems Governance lens to what it terms Ocean Data Science Initiatives, ODSIs, catalogues more than 150 global and regional initiatives built around mobilising ocean data, much of it geospatial and temporal, with the explicit stated goal of improving how oceans are governed rather than simply describing their condition.

The framework's central argument is the part worth internalising. These initiatives, the paper argues, are not passive data providers feeding decisions made

elsewhere. They are creating novel forms of agency and have become significant ocean governance actors in their own right, and in doing so they are blurring what used to be a cleaner distinction between governance architecture, meaning the formal treaties, institutions, and bodies with recognised authority, and governance agency, meaning who actually shapes outcomes in practice.

This is not an abstract academic distinction. It is a direct description of the position HARMONiSE and the Ocean Mapping Hub occupy, or are moving toward, in the Western Indian Ocean. A regional data initiative that determines which coastal zones get prioritised for bathymetric survey, which communities' fisheries data gets structured and shared, and which datasets feed into national marine spatial planning decisions is already shaping those planning outcomes directly, whatever the initiative's own self-description says. The paper's framework gives that role a name, agency, distinct from and sometimes in tension with formal architecture, and a name is useful, because unnamed influence is harder to defend, fund, or hold accountable than influence that is explicitly claimed.

The question is live at the institutional level right now, not just in academic literature. The International Seabed Authority posed a public version of this same question this year, asking whether 2026 is the point at which ocean governance architecture needs formal consolidation to keep pace with exactly the kind of initiative-driven authority this framework describes. That an apex formal institution is asking the question publicly signals the gap between formal architecture and initiative-level agency has become visible enough to name at the highest levels, not just inside specialist governance literature.

The practical implication for HARMONiSE's own positioning is direct. Framing HARMONiSE purely as a data and mapping service, a technical utility sitting neutrally beneath the governance decisions others make, undersells and potentially misdescribes what it actually does once its outputs start determining planning priorities. Framing it explicitly as an ocean data science initiative in this paper's sense, a governance actor exercising a specific, nameable form of agency, is both more accurate and a stronger position from which to argue for a formal seat in

Western Indian Ocean marine spatial planning processes, rather than a support role at their margins.

<https://www.sciencedirect.com/science/article/pii/S258981162200009X>

10

// forty percent of your emissions, and nobody's pricing it right

*Wood Mackenzie just put a number on how much of the world's carbon problem the ocean quietly absorbs for free, and no climate model or carbon price accounts for what happens if that stops.*

#### STORY NOTES

\$ Wood Mackenzie's August 2026 report, Nature Capital: The Unpriced Carbon, puts natural carbon sinks at absorbing roughly 40% of annual human CO<sub>2</sub>

emissions, cutting gross global emissions of 48 gigatonnes down to a net of around 30 gigatonnes reaching the atmosphere. Oceans accounted for approximately 12 gigatonnes of that absorption in 2025 alone, more than double the roughly 6 gigatonnes contributed by forests.

\$ The report's core warning is that these sinks are under increasing stress while the policy and financial frameworks meant to protect them have not kept pace. Compliance carbon markets currently price carbon between US\$5 and US\$90 per tonne, a range the report states does not reflect the true cost of sink degradation, and most energy and climate models do not value natural sinks at all.

\$ Wood Mackenzie modelled outcomes across four scenarios through 2060: a base case, country pledges, net zero,

and delayed transition, none of which currently price the ocean's absorption service as a variable that could itself degrade.

#### STORY ARTICLE

Run the numbers Wood Mackenzie published in August and the ocean is doing roughly a quarter of humanity's entire carbon cleanup job, for free, with no line item anywhere in the global climate finance architecture crediting it for the service. The firm's Nature Capital: The Unpriced Carbon report puts natural carbon sinks, oceans and forests combined, at absorbing approximately 40% of annual human CO<sub>2</sub> emissions, reducing gross global emissions of 48 gigatonnes to a net of roughly 30 gigatonnes actually reaching the atmosphere. Oceans did the larger share of that work in 2025: approximately 12 gigatonnes absorbed,

against roughly 6 gigatonnes from forests.

The report's warning is not that this service might weaken someday. It is that the systems meant to protect and value it have not kept pace with the stress already visible. Compliance carbon markets currently price a tonne of CO<sub>2</sub> somewhere between US\$5 and US\$90 depending on jurisdiction and scheme, and the report states plainly that this range does not reflect what it would actually cost to replace or restore a degraded sink. Most energy and climate models used to plan national and corporate decarbonisation pathways do not value natural sinks as an input at all, treating the ocean's absorption capacity as a fixed background constant rather than a service that itself has a cost of failure.

That fixed-background-constant assumption is the specific error worth naming, because two other stories in this edition already show it breaking down. The AMOC heat valve research covered separately in this issue demonstrates that Atlantic circulation weakening does not just move heat, it changes how much the whole ocean system retains, a mechanism with direct implications for how much carbon the ocean can continue to absorb as circulation patterns shift. Separately, the well-documented 2023 North Atlantic case, where elevated sea surface temperatures reduced CO<sub>2</sub> solubility sharply enough to weaken the regional carbon sink, already showed this is not a distant hypothetical. It happened, in a single ocean basin, inside a single warm year.

Wood Mackenzie modelled its 40% absorption figure across four scenarios through 2060, a base case, country pledges, net zero, and delayed transition, and none of those scenarios appear to treat the natural sink absorption rate itself as a variable that could decline under warming rather than staying fixed at current levels. That is the gap the report's own headline finding points toward without fully closing: if 12 gigatonnes of annual ocean absorption is doing a quarter of the planet's total carbon cleanup, and warming itself measurably degrades that absorption capacity, then every scenario built on today's 40% figure holding steady through 2060 is probably too optimistic, not too conservative.

For any blue carbon or ocean-based climate finance argument, this is the

caution to carry forward from the report rather than the headline statistic alone. Twelve gigatonnes of free annual absorption is an argument for valuing and protecting the service properly. It is not evidence the service is stable enough to keep discounting in the model.

**<https://www.woodmac.com/press-releases/natural-carbon-sinks-absorb-40-of-human-co2-emissions-and-are-not-valued-in-most-climate-models/>**

Morning Edition — compiled 16 September 2026 for Francesca Adrienne, SHORE Institute · Shorelines

Sources: UNFCCC Standing Committee on Finance, Commonwealth Secretariat, GEBCO / Seabed 2030, World Bank, Fitch, Rio Times Online, SPREP / Pacific Islands Roundtable for Nature Conservation, Nature Geoscience / Oregon State University, ScienceDirect (Earth Systems Governance), Wood Mackenzie.